JOURNAL
OF HEPATOLOGY

An overweight female with a history of allergies and drug-induced liver injury

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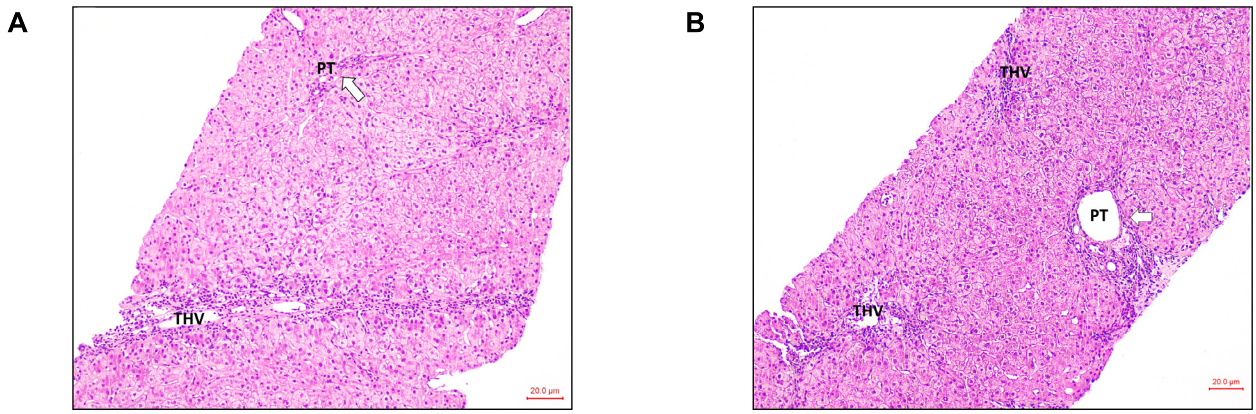

Description

A 47-year-old overweight woman was evaluated for elevated transaminases at her regular checkup. She reported recently taking nimesulide for back pain. Her past medical history included drug-induced liver injury (DILI) 5 years ago, attributed to ampicillin/sulbactam antibiotic therapy, several allergies and a systematic anaphylactic reaction to the flu vaccine. Her BMI was 29 and she admitted social use of alcohol.

Her laboratory results included aspartate aminotransferase (AST) 146 IU/L (upper limit of normal [ULN] 35 IU/L), alanine aminotransferase (ALT) 170 IU/L (ULN 31 IU/L), alkaline phosphatase 50 IU/L (ULN 104 IU/L), gamma-glutamyltransferase 13 IU/L (ULN 36 IU/L), total bilirubin 1.83 mg/dl (ULN 1.0/dl) and γ-globulins 13.5%. Autoantibody serology included ANA (antinuclear antibodies) positivity at 1/160, while AMA (antimitochondrial antibodies), SMA (smooth muscle actin), SLA/LP (soluble liver antigen/liver-pancreas) and LKM (liver kidney microsome) antibodies were negative. Viral hepatitis screening for HAV, HBV, HCV and HEV was negative. Upper abdomen magnetic resonance imaging was unremarkable.

The previous episode of liver enzyme elevation had the characteristics of acute hepatitis and after thorough evaluation and exclusion of any other liver disease had been characterized as DILI. At that time, she was treated with prednisolone 1 mg per kilogram with quick tapering. AST/ALT levels fell but did not normalize completely, fluctuating up to x1.5-2 ULN from time to time. The patient was followed up with regular measurements of liver enzymes, liver ultrasound and vibration-controlled transient elastography without worsening of liver function. She was reluctant to undergo a liver biopsy, so she remained under close follow-up every 2 months for a 6-month period with AST and ALT elevated at x2-3 ULN. During this time liver stiffness increased from 5.8 kPa to 9 kPa and she decided to proceed with the biopsy.

What is your diagnosis?

1) Drug-induced liver injury
2) Alcohol-related liver disease
3) Autoimmune hepatitis
4) Metabolic dysfunction-associated steatotic liver disease

Keywords: autoimmune hepatitis; centrilobular liver injury; drug-induced liver injury.
Received 27 February 2024; received in revised form 29 March 2024; accepted 2 April 2024;
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https://doi.org/10.1016/j.jhep.2024.04.002

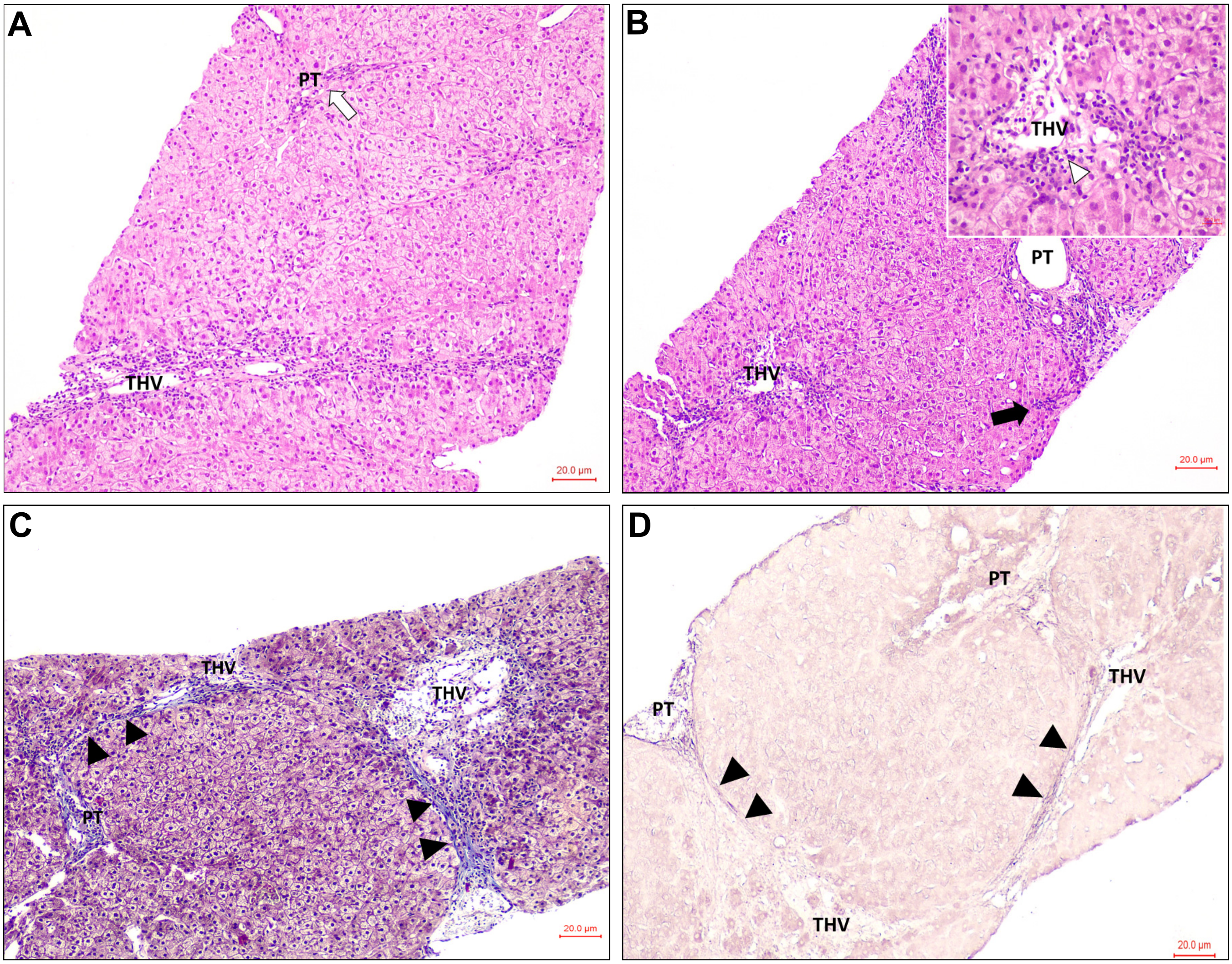

Fig. 1. Liver biopsy showing centrilobular pattern of hepatitic liver injury. (A) Absent portal chronic inflammation and prominent centrilobular inflammation with confluent necrosis around the THV on H&E. White arrow indicates the PT. (B) Another area shows minimal portal inflammation with focal interface activity (black arrow) and confluent necrosis with lymphoplasmacytic inflammation around the terminal hepatic venule on H&E. Inset shows higher magnification of (B), indicating plasma cells (white arrowhead) in the perivenular necroinflammatory infiltrate. (C) Masson trichrome stain shows early portal-central bridging fibrosis (black arrowheads). Collagen fibres stain blue. (D) Orcein stain highlights black elastic fibres in bridging fibrous septa (black arrowheads) indicating chronicity of the fibrotic process. H&E, haematoxylin & eosin, PT, portal tract; THV, terminal hepatic venule.

Diagnosis and outcome

Liver biopsy showed mild distortion of normal tissue architecture with parenchymal extinction and centrilobular reticuline framework collapse, areas of hepatocyte regeneration and few hepatocyte rosettes. Few portal tracts had mild mixed inflammatory infiltrate, including lymphohistiocytes, few plasma cells and few polymorphs with mild interface activity. Interlobular bile ducts were unremarkable and there was mild ductular reaction on keratin 7 (K7) immunostaining. In the parenchyma, the dominant lesion was centrilobular confluent necrosis in many areas and focal central-central bridging necrosis including several plasma cells (Fig. 1A,B). Rare scattered necroinflammatory foci with few apoptotic bodies were also seen. Few K7(+) periportal intermediate hepatocytes indicative of mild chronic cholestasis were noted. Steatosis, ballooned hepatocytes or Mallory-Denk bodies were not present. Masson trichrome stain highlighted mild portal/periportal fibrosis and early central-central bridging fibrosis containing few elastic fibres (orcein stain) (Fig. 1C,D). There were no a1-antirypsin intracytoplasmic globules (PAS-diastase stain), siderosis (Perls stain) or copper-associated granules (orcein stain). Overall, the liver biopsy showed moderate chronic hepatitis with acute exacerbation and prominent centrilobular necrosis (Ishak modified histological activity index total score 8/18) with early bridging fibrosis (Ishak stage 3) and mild chronic cholestasis with no evidence of cholangiopathy. The histological findings were non-specific but they supported an autoimmune aetiology in this patient's clinical context after exclusion of viral and toxic aetiology. The diagnosis of likely autoimmune hepatitis (AIH) with a predominant centrilobular pattern of liver injury was made.

The patient was started on prednisolone 30 mg in combination with azathioprine 50 mg once per day (qd) with quick tapering to 10 mg in a month and liver serology rapidly improved. She is currently stable with normal transaminases under treatment with 5 mg of prednisolone and 100 mg of azathioprine qd. The plan is to next taper out the prednisolone completely and have a stable remission with azathioprine monotherapy. In case of relapse, permanent immunosuppression will be required; in case of stable remission on monotherapy for more than 2 years, a trial of tapering out azathioprine is justified, with or without a prior liver biopsy to confirm remission.[4]

AIH with centrilobular injury

AIH is a heterogeneous disease both clinically and histologically, with clinical presentation ranging from entirely asymptomatic to acute liver failure and liver biopsy findings ranging from non-specific mild chronic hepatitis to severe acute hepatitis with panlobular necrosis. There are no pathognomonic laboratory and pathology findings, but the diagnosis cannot be made without liver biopsy and compatible histology according to most international clinical guidelines. Early diagnosis and treatment with immunosuppressive agents are required to prevent disease progression. Liver histology is part of international AIH diagnostic scoring systems supporting definite, probable or unlikely AIH based on the assumption that AIH has a portal-based pattern of injury. Nevertheless, variable patterns do exist, including the centrilobular-predominant pattern most frequently seen in the context of AIH with acute presentation. Centrilobular hepatitis (also termed central perivenulitis) in AIH usually occurs in association with mainly portal-based chronic inflammation and interface (periportal) hepatitis. However, it may be seen as a predominant lesion in 18-29% or as an exclusive finding in 3% of cases.[1] In some of these cases, autoantibodies or plasma cells may be absent. The centrilobular injury pattern in AIH may represent an early stage of the disease progressing to the more common portal/periportal hepatitis over time. Recent studies indicate that AIH with predominant centrilobular injury may be a distinct AIH subtype with lower serum IgG, lower serum ANA titer and lower relapse rates.[2]

The International AIH Pathology Group recently developed consensus criteria for the histological diagnosis of AIH that can be applied to both acute and chronic presentations of AIH, acknowledging the broad histopathological spectrum of the disease.[3] Using these criteria, the diagnosis of likely, possible, and unlikely AIH can be made based on patterns of portal and lobular inflammation. Using the diagnostic criteria of IAIHPG, cases with predominantly centrilobular injury with minimal or mild portal/periportal inflammation or exclusively centrilobular necrosis can now be diagnosed as AIH and receive life-saving immunosuppression. Studies are ongoing to further highlight which histological criteria are more significant for the differential diagnosis between AIH and DILI in the context of predominantly centrilobular hepatitis.

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Financial support

The authors received no financial support to produce this manuscript.

Conflict of interest

The authors declare no conflict of interest.

Please refer to the accompanying ICMJE disclosure forms for further details.

Authors' contributions

All authors drafted, reviewed, and take final responsibility for the decision to submit the manuscript for publication.

Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhep.2024.04.002.

References

[1] Zhang X, Jain D. The many faces and pathologic diagnostic challenges of autoimmune hepatitis. Hum Pathol 2023;132:114–125. https://pubmed.ncbi.nlm.nih.gov/35753409/.

[2] Ueda K, Aizawa Y, Kinoshita C, et al. Centrilobular zonal necrosis is a unique subtype of autoimmune hepatitis: a cohort study. Medicine (Baltimore) 2022;101(29):e29484. https://pubmed.ncbi.nlm.nih.gov/35866813/.

[3] Lohse AW, Sebode M, Bhathal PS, et al. Consensus recommendations for histological criteria of autoimmune hepatitis from the International AIH Pathology Group. Liver Int 2022;42(5):1058–1069. https://pubmed.ncbi.nlm.nih.gov/35230735/.

[4] Hartl J, Ehlken H, Weiler-Normann C, et al. Patient selection based on treatment duration and liver biochemistry increases success rates after treatment withdrawal in autoimmune hepatitis. J Hepatol 2015;62(3):642–646. https://pubmed.ncbi.nlm.nih.gov/25457202/.